

EHR Workflow Redesign — Business Requirements Document (BRD)

Section 1 — Document Control

i This BRD traces every requirement to a quantified gap from Notebook 2 (Gap Analysis).
Total estimated recovery: AED 9.1M across 6 gaps. Regulatory scope: DHA (Dubai) · DoH/HAAD (Abu Dhabi) · Shafafiya · eSMA.

Version	Date	Author	Changes	Status
0.1	20 May 2026	Amr Thabet	Initial draft — structure and scope	Draft
1.0	27 May 2026	Amr Thabet	Full BRD — all sections complete	Draft

Field	Detail
Document Title	EHR Workflow Redesign BRD
Author	Amr Thabet
Organisation	Leading UAE Health Insurer (Gulf Region)
Status	Draft
Created	20 May 2026
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Section 2 — Executive Summary

i The current EHR-to-insurer claims workflow at a leading UAE health insurer is manual, fragmented, and non-compliant — generating a 33.6% denial rate across 4,500 claims and AED 7.55M in unrecovered revenue, while exposing the organisation to DHA/DoH regulatory risk.

Six gaps have been identified across three parties — Provider, Insurer, and Regulator — each with a quantified AED impact and a regulatory consequence. The gaps span ICD-10 coding errors, missing EHR/RCM integration, manual pre-authorisation, no high-cost member identification, no automated denial follow-up, and absent pre-submission compliance checks.

The proposed solution redesigns the end-to-end workflow from 15 manual steps to 10 automated or semi-automated steps — introducing a real-time ICD-10 validation engine, an EHR-RCM integration layer, an automated pre-auth rules engine, and the P3 predictive model for high-cost member identification (AUC 0.950, catching 87% of high-cost members).

Metric	Current State	Target State
Denial rate	33.6%	Below 15%
Denied value	AED 7,550,894	Recover AED 9.1M
Inpatient denial rate	32.7%	Below 12%
High-cost member spend	AED 45,226,274	Save AED 3,360,000
P3 model AUC	—	0.950
Manual workflow steps	15	10

Section 3 — Scope

This section defines the boundaries of the EHR Workflow Redesign project.

All six gaps identified in Notebook 2 fall within scope. Anything outside these boundaries requires a separate project initiation.

3.1 — In Scope

-  All items below are covered by this BRD and have an assigned owner across Provider, Insurer, or Regulator.

#	In Scope	Party
1	EHR-to-RCM integration layer	Provider
2	Real-time ICD-10 validation engine	Provider
3	Automated pre-authorisation rules engine	Insurer

4	Automated Shafafiya/eSMA claim submission	Insurer
5	Real-time eligibility verification API	Insurer
6	P3 predictive model — high-cost member flagging	Insurer
7	Automated denial routing by reason code	Insurer
8	Pre-submission compliance check engine	All parties
9	DHA/DoH regulatory compliance — ICD-10, pre-auth, 90-day window	Regulator

3.2 — Out of Scope

 The following items have been explicitly excluded. Any future inclusion requires a change request and BRD amendment.

#	Out of Scope	Reason
1	Full EHR system replacement	Capital project — separate initiative
2	Pharmacy and dental claims	Outside current workflow redesign
3	International claims (outside UAE)	DHA/DoH jurisdiction only
4	Patient billing and collections	Provider-side financial system
5	HR or staffing changes	Operational — not within BRD remit

3.3 — Delivery Phases

The redesign is delivered across three phases over nine months, prioritized by regulatory risk and AED recovery value.

Phase	Scope	Timeline
P1 — Foundation	ICD-10 validation + EHR-RCM integration	Months 1–3

P2 — Automation	Pre-auth engine + eligibility API + Shafafiya integration	Months 4–6
P3 — Intelligence	P3 model deployment + denial routing + compliance check	Months 7–9

Section 4 — Stakeholder Register

All stakeholders identified across the three parties — Provider, Insurer, and Regulator — who have a direct interest in or influence over the EHR Workflow Redesign project.

4.1 — Stakeholder Table

-  Influence = decision-making power over project approval or implementation.
Interest = degree to which this stakeholder is affected by the outcome.

#	Role	Party	Responsibility	Interest	Influence
1	Chief Medical Officer	Provider	Clinical workflow sign-off	High	High
2	Revenue Cycle Manager	Provider	EHR-RCM integration owner	High	High
3	Clinical Coder	Provider	ICD-10 compliance	High	Medium
4	Claims Director	Insurer	Pre-auth and denial management	High	High
5	Medical Reviewer	Insurer	Clinical necessity decisions	High	Medium

6	Compliance Officer	Insurer	DHA/DoH regulatory adherence	High	High
7	Data Analyst	Insurer	P3 model deployment and monitoring	Medium	Low
8	DHA Inspector	Regulator	Shafafiya portal compliance	Medium	High
9	DoH/HAAD Auditor	Regulator	ICD-10 and pre-auth enforcement	Medium	High

4.2 — Engagement Strategy

Influence \ Interest	High Interest	Low Interest
High Influence	Manage closely — Claims Director, CMO, Compliance Officer, DHA, DoH	Keep satisfied — N/A
Low Influence	Keep informed — Clinical Coder, Medical Reviewer	Monitor — Data Analyst

i Stakeholders with High Influence and High Interest are the primary sign-off authorities for this BRD. All change requests must be approved by this group before implementation proceeds.

Section 5 — Functional Requirements

Every functional requirement below traces directly to a gap identified in Notebook 2.

No requirement exists without a gap, and no gap is left without a requirement.

5.1 — Requirements Traceability

 Priority levels:

- Critical = regulatory non-compliance risk
- High = direct AED recovery impact
- Medium = operational efficiency gain.

Req #	Gap	Requirement	Priority	Owner	AED Impact
FR-01	Gap 1 — ICD-10 coding errors	System shall validate ICD-10 codes in real-time at point of clinical coding before claim generation	Critical	Provider	AED 1,600,000
FR-02	Gap 2 — No EHR/RCM integration	System shall auto-generate claims from EHR data with no manual re-entry into RCM	High	Provider	AED 1,100,000
FR-03	Gap 3 — Manual pre-auth	System shall apply automated pre-authorisation rules engine — flagging missing pre-auth before processing	Critical	Insurer	AED 1,200,000
FR-04	Gap 4 — No high-cost member ID	System shall run P3 predictive model daily — flagging high-cost members for care management intervention	High	Insurer	AED 3,400,000

FR-05	Gap 5 — No automated denial follow-up	System shall route denied claims automatically by reason code with 90-day resubmission tracker	High	Insurer	AED 400,000
FR-06	Gap 6 — No compliance check	System shall run pre-submission compliance check against DHA/DoH rules before every claim is submitted	Critical	All parties	AED 1,500,000

5.2 — Requirements by Party and Priority

 3 requirements are Critical priority · 3 are High priority.
Combined estimated recovery across all 6 requirements: AED 9,200,000.

Party	Critical	High	Total Requirements
Provider	FR-01	FR-02	2
Insurer	FR-03	FR-04, FR-05	3
All parties	FR-06	—	1

 All Critical requirements must be delivered in Phase 1 or Phase 2.
No sign-off will be granted if any Critical requirement remains unimplemented at go-live.

Section 6 — Non-Functional Requirements

Non-functional requirements define how the system must perform — independent of specific features. All requirements below apply across the full redesigned workflow.

6.1 — Performance

Req #	Requirement	Target	Owner
NFR-01	ICD-10 validation response time	Under 2 seconds per claim	Provider

NFR-02	Eligibility verification API response time	Under 3 seconds per query	Insurer
NFR-03	Pre-auth rules engine processing time	Under 5 seconds per request	Insurer
NFR-04	P3 model daily batch processing	Completed within 2-hour overnight window	Insurer
NFR-05	Shafafiya/eSMA claim submission	Under 10 seconds per claim	Insurer

6.2 — Availability and Reliability

Req #	Requirement	Target	Owner
NFR-06	System uptime — claims processing	99.5% during business hours	Insurer
NFR-07	Planned maintenance window	Outside 7AM–8PM GST	All parties
NFR-08	Disaster recovery — RTO	Recovery within 4 hours	Insurer
NFR-09	Disaster recovery — RPO	Maximum 1 hour data loss	Insurer

6.3 — Security and Compliance

⚠ All systems handling patient data must comply with UAE Federal Law No. 2 of 2019 on the use of Information and Communication Technology in Health Fields and DHA data protection standards.

Req #	Requirement	Target	Owner
NFR-10	Patient data encryption	AES-256 at rest and in transit	All parties
NFR-11	Role-based access control	Applied across all system modules	All parties

NFR-12	Audit trail — all claim actions	Full log retained for minimum 5 years	Insurer
NFR-13	DHA/DoH data residency	All data stored within UAE borders	All parties
NFR-14	Penetration testing	Minimum annually and after major releases	Insurer

6.4 — Usability

Req #	Requirement	Target	Owner
NFR-15	Clinical coder training time	New users productive within 2 days	Provider
NFR-16	System language support	Arabic and English	All parties
NFR-17	Mobile accessibility	Core functions accessible on tablet	Provider

Section 7 — User Stories

-  User stories for this project are managed as Jira tickets.
- 6 Epics — one per gap — with stories and acceptance criteria tracked on the EHR Workflow Redesign board.

Section 8 — Success Metrics and KPIs

Every KPI below traces directly to a gap closed by this redesign. Baseline values are derived from P1 and P3 analysis. Targets represent the expected state 12 months post-implementation.

8.1 — Claims and Denial KPIs

KPI	Baseline	Target	Gap Closed	Owner
Overall denial rate	33.6%	Below 15%	Gap 1, 2, 3, 6	Insurer

Denied claim value	AED 7,550,894	Reduce by 70%	Gap 1, 2, 3, 6	Insurer
Inpatient denial rate	32.7%	Below 12%	Gap 3	Insurer
ICD-10 coding error rate	Not measured — errors only discovered at rejection	Below 2%	Gap 1	Provider
Pre-auth missing rate	Embedded in 32.7% inpatient denial rate	0%	Gap 3	Insurer
Claims submitted within 90 days	No tracking system in place — window breaches undetected	100%	Gap 6	All parties

8.2 — Financial Recovery KPIs

Gap	Solution	Estimated Annual Recovery	Priority
Gap 1 — ICD-10 coding errors	Real-time validation engine	AED 1,600,000	Critical
Gap 2 — No EHR/RCM integration	Automated claim generation	AED 1,100,000	High
Gap 3 — Manual pre-auth	Automated rules engine	AED 1,200,000	Critical
Gap 4 — No high-cost member ID	P3 predictive model daily run	AED 3,400,000	High
Gap 5 — No denial follow-up	Automated routing + 90-day tracker	AED 400,000	High
Gap 6 — No compliance check	Pre-submission compliance engine	AED 1,500,000	Critical
Total	All 6 gaps closed	AED 9,200,000	—

i Recovery values represent currently unrecovered revenue identified in Notebooks 1 and 2. Full recovery is projected within 12 months of complete implementation across all three

delivery phases.

8.3 — Model and System KPIs

KPI	Baseline	Target	Owner
P3 model AUC	0.950	Maintain above 0.920	Data Analyst
High-cost member detection rate	87%	Maintain above 85%	Data Analyst
ICD-10 validation response time	Manual — no validation in place	Under 2 seconds	Provider
Pre-auth processing time	Manual — 3 to 5 business days	Under 5 seconds	Insurer
Claim resubmission rate	0% — no tracking system exists	Above 90% within 90 days	Insurer
System uptime	No integrated system — fully manual	99.5% during business hours	Insurer

⚠️ KPIs in sections 8.1 and 8.3 must be reviewed monthly for the first 6 months post-implementation. Any KPI falling below target triggers an immediate review by the Claims Director and Compliance Officer.

Section 9 — Assumptions and Constraints

This section documents the assumptions made during BRD development and the constraints that bound the solution. Any assumption that proves false or any constraint that changes must trigger a formal BRD review.

9.1 — Assumptions

ℹ️ All requirements and KPI targets in this BRD are contingent on the assumptions below holding true at the time of implementation.

#	Assumption	Party	Impact if False
A-01	DHA/DoH regulatory requirements remain unchanged during implementation	Regulator	Full BRD review required

A-02	Provider EHR system supports API integration with RCM	Provider	Gap 2 solution requires redesign
A-03	Shafafiya and eSMA portals support automated API submission	Regulator	Gap 6 solution requires redesign
A-04	P3 predictive model maintains AUC above 0.920 in production	Insurer	Model retraining required before deployment
A-05	Claims data quality remains consistent with P1 training dataset	Insurer	Model performance will degrade
A-06	All three parties commit named resources to each delivery phase	All parties	Phase timelines will slip
A-07	Budget is approved for all three delivery phases before Phase 1 begins	Insurer	Phased delivery at risk

9.2 — Constraints

Constraints are fixed boundaries that cannot be changed within the scope of this project. Any constraint requiring modification must be escalated to the sign-off authority group.

#	Constraint	Type	Impact
C-01	All systems must comply with UAE Federal Law No. 2 of 2019 on ICT in Health	Regulatory	Non-negotiable — no exceptions
C-02	All patient data must be stored within UAE borders	Regulatory	Rules out non-UAE cloud providers
C-03	Claims must be submitted via Shafafiya portal — no alternative channel	Regulatory	Integration must target Shafafiya API
C-04	ICD-10 coding is mandatory for all claims — no legacy coding accepted	Regulatory	Validation engine must use ICD-10 only
C-05	90-day submission window is a hard DHA/DoH deadline — no extensions	Regulatory	Tracker must alert at 60 days minimum
C-06	Full EHR system replacement is out of scope	Project	Solution must work with existing EHR

C-07	Implementation budget is fixed — no scope creep without change request	Financial	All phases must deliver within agreed budget
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Section 10 — Sign-Off

This section records the formal approval of the Business Requirements Document by all sign-off authorities. No implementation work may begin until all Critical and High Influence stakeholders have signed off.

 This BRD is a controlled document. Any changes after sign-off require a formal change request approved by all parties listed below. Version history is maintained in Section 1.

10.1 — Sign-Off Authority

Role	Signature	Date
Chief Medical Officer	—	—
Revenue Cycle Manager	—	—
Claims Director	—	—
Compliance Officer	—	—
DHA Inspector	—	—
DoH/HAAD Auditor	—	—

10.2 — BRD Status

Field	Detail
Current status	Draft — pending stakeholder review
Submitted for review	27 May 2026
Target sign-off date	27 June 2026
Next scheduled review	27 June 2026
Version	1.0

Upon full sign-off this document will be updated to Status: Approved and locked for editing. All subsequent changes must follow the change request process documented in Section 9.2 — Constraint C-07.